

Setup and networks

Finish everything here before Day 2: tools, accounts, a testnet, test tokens, and the starter running on your laptop. Installs and faucet waits are the most common way teams lose their build morning, so split the list across teammates.

github.com/murthyroshan/innoblock-2.0-starter

On every laptop

- ☐ Chrome, Brave or Edge with the **MetaMask** extension (metamask.io)
- ☐ MetaMask → Settings → Advanced → turn on **Show test networks**
- ☐ A **new wallet just for the hackathon** (a "burner"). Never import a wallet that has held real money
- ☐ **Python 3.10 or newer:** `python --version`
- ☐ **Git:** `git --version`
- ☐ **VS Code** (or any editor)

Accounts (one per team is enough, all free)

Service	What for
GitHub	Your repo and your submission
Render	Hosts the Flask backend
Vercel or Netlify	Hosts the frontend
Neon or Supabase	Free Postgres, so deployed records survive restarts
UptimeRobot	Keeps the free backend awake
An AI provider	An API key, only if your idea uses AI

Test tokens

- ☐ Pick your network ([Networks and faucets](#)). If unsure: **Ethereum Sepolia**
- ☐ Every wallet that sends transactions holds test tokens, **including the backend's wallet**. Around 0.05 Sepolia ETH (or 0.5 POL on Amoy) is plenty for a whole day of testing
- ☐ One teammate who gets tokens can share them: MetaMask → Send → paste address. It arrives in seconds

Clone the starter

```
git clone https://github.com/murthyroshan/innoblock-2.0-starter.git
cd innoblock-2.0-starter/backend
python -m venv venv
venv\Scripts\activate          # Mac/Linux: source venv/bin/activate
pip install -r requirements.txt
```

If `pip install` works tonight, it will work tomorrow.

10-minute self-test

1. Open the contract in Remix with this [one-click link](#).
2. **Solidity compiler** tab → **Compile RecordRegistry.sol**.
3. **Deploy & run transactions** tab → Environment: **Browser Extension**, then **MetaMask** → approve the connection → check MetaMask shows your testnet.
4. **Deploy** → confirm in MetaMask.
5. Open the transaction from MetaMask's **Activity** tab on the block explorer.

If you see your contract creation on the explorer, your laptop and wallet are ready. Keep the address: you can use this deployment tomorrow. Screenshots of each Remix step are in [02 · Build](#).

Networks and faucets

Any public **testnet** is allowed. The starter supports five EVM testnets out of the box. If you're unsure, use **Ethereum Sepolia**: it has the most faucets, tutorials and mentor experience.

Supported networks

Network	Chain ID (hex)	Currency	Public RPC	Explorer
Ethereum Sepolia	11155111 (0xaa36a7)	ETH	https://ethereum-sepolia-rpc.publicnode.com	sepolia.etherscan.io
Base Sepolia	84532 (0x14a34)	ETH	https://base-sepolia-rpc.publicnode.com	sepolia.basescan.org
Polygon Amoy	80002 (0x13882)	POL	https://polygon-amoy-bor-rpc.publicnode.com	amoy.polygonscan.com
Arbitrum Sepolia	421614 (0x66eee)	ETH	https://arbitrum-sepolia-rpc.publicnode.com	sepolia.arbiscan.io
OP Sepolia	11155420 (0xaa37dc)	ETH	https://optimism-sepolia-rpc.publicnode.com	sepolia-optimism.etherscan.io

Don't use Goerli or Mumbai. Goerli was shut down in April 2024. Polygon Mumbai was deprecated on 13 April 2024 and replaced by **Polygon Amoy**. Old tutorials still mention both.

Switching the starter to another network

The same contract works on every network. Change three places:

1. **Remix**: switch MetaMask to the network, then deploy `RecordRegistry.sol` again. Each network gets its own contract address.
2. **backend/.env**: set `RPC_URL`, `CONTRACT_ADDRESS` and `EXPLORER_URL` for that network (examples are in `.env.example`).
3. **frontend/config.js**: set `ACTIVE_NETWORK` (for example `"baseSepolia"`) and `CONTRACT_ADDRESS`.

The frontend asks MetaMask to switch network automatically, and adds the network to MetaMask if it isn't there yet.

Any other EVM testnet

Add an entry to `NETWORKS` in `frontend/config.js` with its name, chain ID, RPC URL, explorer and currency. Look up the details on chainlist.org (tick "Include testnets").

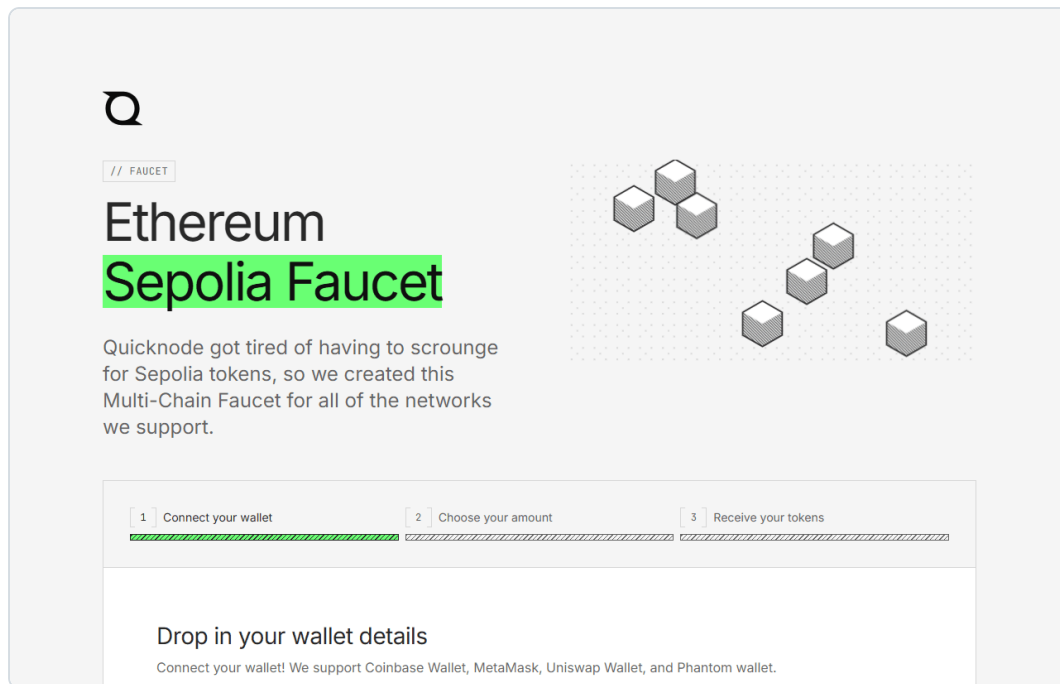
Non-EVM chains

Other testnets (Solana devnet and so on) are allowed, but this starter and its guides only cover EVM chains.

Faucets

Many faucets now ask for a small **real** balance on Ethereum mainnet, which a new hackathon wallet doesn't have. These ones work without it:

Network	Faucets that work with a new wallet
Ethereum Sepolia	QuickNode (no account) · Google Cloud (Google account) · PoW faucet (no account; you "mine" in a browser tab, so leave it running)
Base Sepolia	QuickNode · Coinbase Developer Platform (free developer account)
Polygon Amoy	QuickNode · Polygon faucet
Arbitrum Sepolia	QuickNode
OP Sepolia	QuickNode



QuickNode gives one claim every 12 hours per wallet per network. **If every faucet fails**, ask a teammate or another team to send you some (MetaMask → Send), or ask the organisers on Day 1, not on Day 2. Faucet rules change often; these were checked on 4 October 2026.

Wallet

Use **MetaMask**: the starter and every guide assume it. Turn on **Show test networks** in Settings → Advanced.

One wallet extension at a time. If several are installed they fight over `window.ethereum` and the wrong popup opens. Disable the others before you demo.

On a phone, `window.ethereum` only exists inside a wallet's built-in browser. Open your app's URL from the MetaMask app's browser tab.